



# Suraj Ganachari

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Portfolio: [suraj-ganachari.github.io/atelier/](https://suraj-ganachari.github.io/atelier/)

## EDUCATION

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### Bachelor of Science, Mechanical Engineering Honors, 08/2025 – 05/2029

The University of Texas at Austin

- Overall GPA: **4.00/4.00**
- Activities & Societies: Longhorn Baja Racing, Longhorn Design Lab, ASME

## EXPERIENCE

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### Mechanical Engineering Intern, ERock, 05/2026 – 08/2026

- Collaborated with R&D to design a natural gas engine test cell, managing system requirements, BOMs, and sourcing.
- Executed physical modifications to generator enclosures for new filters and reconfigured the control panel wiring.
- Authored detailed Standard Operating Procedures (SOPs) and executed multiple hands-on R&D engineering projects.
- Gained exposure to the full product lifecycle, spanning from initial design to Factory Acceptance Testing (FAT).

### Powertrain Racing Team Member, Longhorn Baja Racing, 09/2025 – Present

- Designing gears and casing for a custom two-stage, twin-output reduction gearbox with direct-drive rear axles.
- Researching COTS and custom gearbox feasibility, analyzing optimal gear ratios for UT's Baja SAE competition car.
- Authoring collaborative powertrain research paper, presenting findings on gearbox design and integration.

### Research Assistant, UT Freshman Introduction to Research in Engineering (FIRE), 01/2026 – 04/2026

- Optimizing Dr. Guihua Yu's atmospheric water collector designs to actively bring potable water to dry climate locations.
- Engineering a hydrogel cartridge support system for commercial prototypes utilizing Solidworks and FDM printing.
- Evaluating device performance parameters to refine designs, culminating in a formal research report and presentation.

## PROJECTS

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### Personal 1/10 Scale RC Car Project, 12/2025 – Present

- Building a 1/10 scale RC car from the ground up as a solo project to learn core vehicle systems and fabrication skills.
- Designing and 3D printing custom chassis, steering, suspension, and drivetrain components, including a differential.
- Integrating motor, ESC, battery, and ESP-NOW control electronics while modeling a custom body in Autodesk Alias.

### Odor-Reducing Laundry Hamper, 10/2025 – Present

- Researching moisture, bacteria, & VOC reduction methods for a specialized filtration system within a laundry hamper.
- Developing a functional proof-of-concept for a formal pitch to an investment firm regarding future product viability.

## LEADERSHIP & MENTORSHIP

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### First-Year Interest Group (FIG) Mentor, UT First-Year Experience Office, 01/2026 – Present

- Facilitating a seminar for incoming engineering students to aid their academic and social college transition.
- Mentoring a cohort of engineering freshmen on academic success strategies and key university resource access.

## SKILLS

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Software & Tools: Onshape, Solidworks, Fusion, Alias, AutoCAD, Google Workspace, Microsoft Office

Programming: Python, C++ (working knowledge)

## ACCOMPLISHMENTS

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### Virginia & Ernest Cockrell, Jr. Scholarship in Engineering

“Most Technical Design” award at UT Makeathon (1/18 teams)